Supplementary Material

Directional transfer of chemical-gene effects from rodents to humans: an independence-guarded meta-analysis of the Comparative Toxicogenomics Database

This file contains the supplementary tables referenced in the main text. Unless stated otherwise, the analytic population is the primary set: chemical-gene-endpoint effect keys measured in both a human and a rodent context, with an unambiguous consensus direction in each species, and with disjoint supporting PubMed identifiers across species (n=40,779). Agreement is the probability that the human consensus direction equals the rodent consensus direction. Point estimates carry Wilson 95% intervals; interval estimates labelled "bootstrap" resample chemicals (1,000 replications) to respect the nesting of effect keys within chemicals. All six figures referenced in this supplement appear in the main text (Figures 1-6).

Contents. The four headline comparisons (overall independent agreement, all pairs, shared-publication pairs, and the mouse-versus-rat within-rodent benchmark), with both Wilson and cluster-bootstrap intervals, are given in Table 1 of the main text and are not repeated here. The tables below provide the stratified and mechanistic detail that supports the main text.

S1. Stratified reproducibility

Table S1: Independent human-rodent directional agreement by molecular endpoint (primary set), ordered by sample size. Endpoints with fewer than 30 pairs are omitted. Methylation agreement is indistinguishable from its chance baseline.

Molecular endpoint	n	Agreement	Bootstrap 95% CI	Chance
Expression	36,986	60.8%	58.3-63.6	50.4%
Activity	1,106	89.3%	87.2-91.2	50.8%
Methylation	1,102	41.9%	37.1-64.4	40.9%
Phosphorylation	744	81.5%	77.5-85.0	64.0%
Secretion	286	85.7%	81.1-90.6	75.1%
Cleavage	182	96.7%	94.2-98.9	95.7%
Response to substance	144	75.0%	67.3-81.9	50.3%
Metabolic processing	55	98.2%	94.2-100	98.2%
Abundance	53	84.9%	73.2-95.4	71.2%
Uptake	44	95.5%	89.1-100	95.5%
Transport	40	92.5%	80.5-100	92.7%

Table S2: Independent directional agreement by replication depth, the minimum number of supporting publications per species. Agreement rises monotonically to determinism.

Minimum supporting reports per species	n	Agreement	Bootstrap 95% CI
1	38,135	60.7%	57.6-63.7
2	1,980	76.5%	70.1-84.9
3	374	90.6%	85.7-96.8
4	135	95.6%	92.6-98.4
5	60	100%	100-100
$\geq$6	95	100%	100-100

Table S3: Independent directional agreement by coarse chemical class (MeSH chemical tree). The spread across classes is small, consistent with the near-zero chemical-class variance component in Table S5.

Chemical class	n	Agreement	Bootstrap 95% CI
Heterocyclic organic	4,637	68.6%	63.3-74.3
Other	15,263	63.3%	60.3-66.4
Solvents / industrial	2,673	61.7%	58.4-69.6
Organic pollutants	13,853	59.3%	54.2-67.8
Metals / inorganic	4,336	59.1%	53.5-70.1

S2. Decomposition of non-agreement

Table S4: Within-species directional inconsistency. For keys curated more than once within a single species, the fraction that were internally ambiguous (the same chemical-gene-endpoint curated both up and down). This is a within-species reproducibility deficit that is present before any cross-species comparison.

Species	Keys curated	Keys curated >1 time	Internally ambiguous
Human	580,801	37,780	58.3%
Mouse	380,904	30,685	54.5%
Rat	299,647	22,840	65.9%

Table S5: Variance decomposition of the binary agreement outcome (primary set), as one-way intraclass correlation coefficients (ICC) by grouping factor. The ICC is the share of agreement variance attributable to differences between groups. Transferability localises to the specific chemical-gene pair; chemical class explains almost none of the variance.

Grouping factor	Groups	ICC
Chemical-gene pair	40,200	0.442
Molecular endpoint	17	0.117
Gene	11,949	0.068
Chemical	1,477	0.046
Chemical class	7	0.005

Human-specific divergence (main text, Figure 4A): human-versus-rodent directional disagreement was 38.0% against a mouse-versus-rat disagreement ceiling of 33.8%, an excess of 4.2 percentage points (cluster-bootstrap 95% CI -1.1 to 10.9), not distinguishable from zero.

S3. Predictive modelling

Table S6: Predicting directional transfer for unseen chemicals under grouped (by chemical) five-fold cross-validation, so that no chemical appears in both training and test folds. AUC, area under the ROC curve; AP, average precision; Brier, mean squared error of the predicted probability. The prevalence baseline predicts the overall agreement rate (62.0%) for every effect.

Model	AUC (mean ± SD)	AP	Brier
Gradient boosting	0.583 ± 0.025	0.725	0.231
Logistic regression	0.562 ± 0.014	0.712	0.230
Prevalence baseline	0.500 ± 0.000	0.620	0.236

Permutation importance (gradient boosting; AUC drop when feature permuted)

Feature	AUC drop
Rodent consensus direction	0.056
Gene promiscuity (log)	0.037
Molecular endpoint	0.021
Chemical promiscuity (log)	0.016
Rodent supporting publications	0.013
Rodent support (vote count)	0.000
Gene pathway degree (log)	−0.001
Chemical class	−0.009

Leave-one-chemical-class-out generalisation (gradient boosting)

Held-out class	n	AUC	Observed agreement
Solvents / industrial	2,673	0.610	61.7%
Heterocyclic organic	4,637	0.604	68.6%
Metals / inorganic	4,336	0.571	59.1%
Other	15,263	0.564	63.3%
Organic pollutants	13,853	0.511	59.3%

S4. External conservation anchor

The anchor was pre-registered before modelling: human-rodent protein percent identity (Ensembl Compara, per-gene mean across mouse and rat) and one-to-one orthology (Ensembl and NCBI `gene_orthologs`), all external to CTD. Coverage of the primary set was near-complete: 39,933 of 40,779 keys (97.9%) carried an Ensembl identity value (median 87.5%).

Table S7: Independent directional agreement by quartile of human-rodent protein identity. Agreement is flat across the full identity range.

Identity quartile	Median identity	n	Agreement	Bootstrap 95% CI
Q1 (lowest)	70.4%	9,987	63.0%	60.6-66.2
Q2	83.4%	9,987	61.8%	58.5-65.2
Q3	90.5%	9,976	62.9%	59.9-66.1
Q4 (highest)	96.7%	9,983	60.5%	56.5-64.4

Table S8: External-anchor association with directional transfer. The trend is a chemical-cluster-robust logistic regression of agreement on protein identity; the orthology rows are odds ratios contrasting one-to-one orthologs against all others. The incremental rows compare the Aim 3 gradient-boosting model with and without the two conservation features under the same grouped cross-validation, restricted to keys with a conservation value. All anchors are null.

Anchor / model	Estimate (95% CI)	p
Protein identity, per +10 percentage points	OR 0.97 (0.94-1.00)	0.07
Ensembl one-to-one orthology	OR 1.02 (0.96-1.09)	0.49
NCBI one-to-one orthology	OR 0.92 (0.78-1.09)	0.33
CTD-internal model	AUC 0.574 (SD 0.030)	–
CTD-internal + conservation	AUC 0.576 (SD 0.023)	–
Difference (paired across folds)	ΔAUC +0.002 (SD 0.009)	–

S5. Catalogue of independent cross-species conflicts

Table S9: Effect keys that are independently and repeatedly curated in *opposite* directions in humans and rodents (disjoint supporting publications, unambiguous consensus in each species). These are candidate true species divergences or curation errors for targeted review. The direction column gives the human then rodent consensus; "n PMIDs" gives supporting publications on each side. A representative selection is shown; the complete catalogue is distributed with the analysis code.

Chemical	Gene	Endpoint	Human / rodent	Human PMIDs	Rodent PMIDs
Bisphenol A	RPL5	expression	decrease / increase	4	5
Paraquat	GCLC	expression	decrease / increase	2	7
Dioxin (TCDD)	ALPL	expression	increase / decrease	2	7
Dioxin (TCDD)	IL33	expression	increase / decrease	2	7
Bisphenol A	ARL3	expression	increase / decrease	2	6
Bisphenol A	RPL22L1	expression	decrease / increase	3	5
Bisphenol A	RPL9	expression	decrease / increase	2	6
Resveratrol	IGF1R	expression	decrease / increase	4	4
Silicon dioxide	LCN2	expression	decrease / increase	3	6
Bisphenol A	RPL13A	expression	decrease / increase	3	5
Bisphenol A	RPL7	expression	decrease / increase	2	4
Cadmium chloride	CTNNB1	expression	increase / decrease	2	5
Dioxin (TCDD)	IGFBP6	expression	decrease / increase	3	4
MPP^+	LDHA	expression	increase / decrease	1	5
4-Nonylphenol	BAX	expression	decrease / increase	2	4
Arsenite	IL6	expression	decrease / increase	3	4
Bisphenol A	MYO18A	expression	increase / decrease	2	4
Bisphenol A	TBRG4	expression	decrease / increase	3	3

The catalogue is enriched for intensively studied environmental chemicals (bisphenol A, dioxin, paraquat, cadmium) acting on ribosomal and stress-response genes, the regime in which high-throughput expression profiling produces the most context-dependent directional calls.